

B-JEPA: A Comprehensive Literature Review for Bayesian Joint Embedding Predictive Architecture in Gene Regulatory Network Inference

B-JEPA sits at an unoccupied intersection of three mature fields — JEPA-style latent prediction, Bayesian sparse inference with horseshoe priors, and gene regulatory network reconstruction — combining ideas that no single existing method or obvious pairing of methods has brought together. The architecture treats GRN inference as a predictive coding problem in learned latent space, where transcription factor expression profiles serve as context, downstream gene expression serves as prediction targets, and horseshoe priors enforce biologically motivated sparsity while providing calibrated posterior distributions over regulatory edges. This review covers twenty papers across five clusters that collectively define B-JEPA's intellectual foundations, direct competitors, statistical machinery, theoretical framing, and validation ecosystem. The analysis reveals that while each component of B-JEPA has precedent, their integration creates genuine methodological novelty addressing a gap explicitly identified by the virtual cell community: no existing method combines latent representation learning, Bayesian sparse inference, and calibrated uncertainty quantification for gene regulatory network reconstruction.

Cluster 1: The JEPA paradigm and its theoretical foundations

LeCun's cognitive architecture establishes prediction in latent space as the core principle

LeCun (2022) proposed a complete cognitive architecture for autonomous intelligent agents in a 62-page position paper that introduced JEPA as the central world model component. (OpenReview) The architecture comprises six differentiable modules — a perception module, world model, cost module, actor, short-term memory, and configurator — designed to support both reactive (System 1) and deliberative (System 2) behavior. (openreview) (OpenReview) The critical architectural innovation is that JEPA predicts in **abstract representation space** rather than input space. (Emergent Mind) The energy function takes the form $E(x, y, z) = D(\text{Pred}(\text{Enc}(x), z), \text{Enc}(y))$, where separate encoders map inputs to latent representations and a predictor, conditioned on latent variable z , maps one representation to another.

LeCun's argument against generative models is that they waste capacity predicting irrelevant pixel-level detail — exact leaf positions, lighting reflections — that is fundamentally unpredictable. JEPA's encoders discard such detail, allowing the predictor to operate over semantically meaningful abstractions. (arxiv +2) The argument against contrastive methods is that they learn only to distinguish compatible from incompatible pairs without learning to predict or fill in missing information. (arxiv) (Rohit Bandaru) The latent variable z is particularly

important: it parameterizes the space of plausible predictions, enabling multimodal outcomes without enumerating all possibilities in input space. (openreview +2)

For B-JEPA, this paper provides the foundational blueprint. The x-encoder processes TF expression profiles, the predictor (conditioned on latent z) predicts target gene embeddings, and the y-encoder encodes actual target gene expression. Crucially, **LeCun's latent variable z maps directly to B-JEPA's Bayesian posterior over regulatory weights** — uncertainty about which regulatory relationships exist is encoded as a distribution over z rather than a point estimate. The paper is cited as the seminal reference establishing why prediction in learned representation spaces, rather than input reconstruction, is the principled approach.

I-JEPA demonstrates the architecture works without hand-crafted augmentations

Assran et al. (2023) instantiated JEPA for images, achieving **79.3% top-1 accuracy** on ImageNet-1K linear evaluation with ViT-H/14 — competitive with augmentation-heavy methods like iBOT — while using no hand-crafted data augmentations whatsoever. (arxiv +2) The architecture uses a context encoder (ViT processing visible patches), a target encoder (EMA of the context encoder processing full images), and a narrow transformer predictor conditioned on positional mask tokens. (TheCVF) Four target blocks (scale 0.15–0.2 of image area) are masked, and the predictor must infer their representations from a single large context block (scale 0.85–1.0). (arxiv)

The masking strategy is the key design choice: target blocks must be large enough to require semantic understanding (not just texture interpolation), while the context block must be spatially informative but incomplete. (arXiv) (Meta) Collapse is prevented through the asymmetric EMA target encoder — no negative samples, no contrastive loss needed. The predictor architecture is deliberately narrow (fewer layers, smaller dimension than the encoder), forcing it to learn abstract mappings rather than memorizing identity functions. (arXiv) Training is **2.5× faster** than comparable methods and **10× more efficient** than MAE at the same model scale.

(arXiv) (arXiv)

For B-JEPA, I-JEPA provides the most direct architectural template. The context encoder processes TF expression profiles (analogous to visible patches), the predictor maps these to target gene expression embeddings (analogous to masked patch representations), and the target encoder provides ground-truth embeddings. The natural information asymmetry in gene regulation — TFs regulate target genes, creating an inherent "masking" — replaces the artificial spatial masking of I-JEPA. B-JEPA's predictor is conditioned on regulatory graph structure (which TFs regulate which genes) rather than spatial positions, adapting I-JEPA's positional conditioning to biological topology.

V-JEPA extends latent prediction to temporal and sequential data

Bardes et al. (2024) demonstrated that JEPA handles temporal prediction effectively, (arXiv) achieving **81.9%** on Kinetics-400 and **72.2%** on Something-Something-v2 (Meta) (a **+10 point** improvement over previous best) under frozen evaluation — no fine-tuning of the pretrained

encoder. (arXiv) (Meta) The architecture extends I-JEPA to spatio-temporal data with ~90% masking ratio across both spatial and temporal dimensions. (Emergent Mind) Large temporal masks force the model to learn meaningful dynamics rather than trivial interpolation between adjacent frames. (Meta)

The temporal prediction paradigm maps directly to B-JEPA's causal inference structure. In V-JEPA, past frames (context) predict future frame representations (targets); in B-JEPA, TF activity profiles (upstream cause) predict downstream target gene expression representations (effect). The "temporal dimension" in gene regulation corresponds to the causal chain: TF binds promoter → transcription → mRNA → protein → phenotypic effect. V-JEPA's high masking ratio (~90%) mirrors GRN sparsity — most TF-gene pairs have no regulatory relationship. The later V-JEPA 2 addition of action-conditioned prediction further strengthens this analogy: (arXiv) (arXiv) in B-JEPA, the "action" is the TF activity level conditioning the predictor.

LeJEPA proves isotropic Gaussian embeddings are optimal, grounding B-JEPA's prior choice

Balestrieri and LeCun (2025) established the first provably optimal JEPA by showing that **isotropic Gaussian embedding distributions minimize worst-case downstream prediction risk** across all possible linear and nonlinear tasks. (arxiv) The proof rests on two axioms: solve the prediction task (embedding agreement between views) and enforce isotropic Gaussian embeddings. Under OLS and ridge regression, the isotropic Gaussian uniquely minimizes worst-case bias and variance; for k-NN and kernel regression, it maximizes entropy among fixed-variance distributions, ensuring maximal representational spread. (emergentmind)

The practical contribution is SIGReg (Standardize-Isotropize-Gaussianize Regularization), which uses the Cramér-Wold theorem to enforce Gaussianity by matching 1D marginal projections to $N(0,1)$ via the Epps-Pulley characteristic function test. (Le-wm) This scales as $O(N \cdot K)$ — linear in both batch size and embedding dimension — and eliminates the need for stop-gradients, EMA, or teacher-student architectures. (arxiv) LeJEPA achieves **79% linear probe accuracy** with ViT-H/14 on ImageNet (emergentmind) and demonstrates a **94.52% Spearman correlation** between training loss and downstream accuracy — the first JEPA where the training objective reliably predicts transfer performance. (arxiv) Notably, in-domain pretraining on the small Galaxy10 astronomy dataset (11K samples) outperformed DINOv2 transfer from ImageNet, demonstrating the viability of domain-specific pretraining on limited data.

(emergentmind)

This is arguably the most theoretically important paper for B-JEPA. The proof that isotropic Gaussian embeddings are optimal provides direct justification for placing Gaussian priors on B-JEPA's latent gene expression representations. SIGReg could be incorporated into B-JEPA's training objective to ensure well-structured embedding spaces. The Galaxy10 result is directly relevant: B-JEPA operates on domain-specific genomic data of limited scale, and LeJEPA demonstrates this is not a barrier to learning useful representations.

Synthesis: gaps in the JEPA literature that B-JEPA addresses

The JEPA paradigm is architecturally mature but has six critical gaps. **No JEPA has been applied to biological or genomic data** — all work remains in computer vision. **No JEPA incorporates Bayesian uncertainty quantification** — all produce point estimates without posterior distributions, credible intervals, or principled model comparison. **No JEPA handles structured sparse prediction** — existing JEPAs predict dense representations (all patches/frames), while B-JEPA must infer a sparse graph where most edges are zero. **No JEPA conditions on regulatory or causal graph structure** — I-JEPA conditions on spatial positions, V-JEPA on spatio-temporal positions, but B-JEPA conditions on which TFs are known or hypothesized regulators of which genes. **No JEPA produces interpretable edge-level outputs** — current JEPAs optimize for representation quality probed post-hoc, while B-JEPA must produce a weighted, signed regulatory network with per-edge uncertainty. Finally, while LeJEPA proves Gaussian optimality generically, **no work has connected this to domain-specific biological priors** like the known sparsity of gene regulatory networks.

Cluster 2: Bayesian and frequentist methods for gene regulatory network inference

DREAM5 defines the benchmark landscape and reveals that no single method dominates

Marbach et al. (2012) conducted the most comprehensive blind assessment of GRN inference methods, comparing **35 approaches** across four expression datasets. (Semantic Scholar) (nih) The E. coli dataset — **806 microarray experiments, 4,511 genes, 334 TFs, and 67 TF-specific knockout experiments** — with a gold standard curated from RegulonDB became the canonical benchmark for the field. (Broadinstitute) Methods were evaluated using AUROC and AUPR, with rank-based integration across datasets. (PubMed Central)

The results were striking in their heterogeneity. No single method dominated all datasets. (nih) Regression methods (TIGRESS, GENIE3) generally performed well individually, while **Bayesian network methods ranked among the worst** — a cautionary finding for any Bayesian approach. (PubMed Central) However, community/ensemble integration via Borda count was robustly the best strategy, achieving the highest overall performance. (nih) Methods exploiting knockout information significantly outperformed those that did not. (PubMed Central) Experimental validation of 53 novel E. coli predictions yielded a **43% confirmation rate**, (nih) establishing a realistic baseline for discovery. The yeast dataset proved nearly intractable, with most methods barely exceeding random performance — suggesting that eukaryotic GRN inference requires richer data types than bulk microarray alone.

For B-JEPA, DREAM5 serves as the primary benchmark. The E. coli dataset's properties — moderate sample size, known perturbation structure, well-curated gold standard — make it the standard test bed for comparing AUROC and AUPR against the established leaderboard. The poor performance of classical Bayesian network methods in DREAM5 creates both a challenge

and an opportunity: B-JEPA must demonstrate that its modern Bayesian approach (horseshoe priors, JEPA representations) overcomes the limitations of earlier Bayesian methods (combinatorial DAG search, weak priors).

BiGSM represents the most directly comparable recent Bayesian approach

Qin et al. (2025) developed BiGSM (Bayesian Inference of Gene Regulatory Network via Sparse Modelling), the most recent Bayesian GRN inference method (nih) and B-JEPA's closest methodological neighbor. (Oxford Academic) BiGSM assumes a linear steady-state model $Y = AY + P + \varepsilon$, where A is the sparse GRN adjacency matrix and P is a known perturbation matrix. Sparsity is induced through Sparse Bayesian Learning (SBL): independent Gaussian priors $N(0, \alpha_{ij}^{-1})$ are placed on each edge weight, with precision hyperparameters α_{ij} learned via Type-II maximum likelihood. As learning progresses, many $\alpha_{ij} \rightarrow \infty$, forcing corresponding weights to zero through automatic relevance determination.

Because the prior is conjugate Gaussian, the posterior over each row of A is analytically tractable — a **closed-form multivariate Gaussian** with computable mean and covariance. BiGSM achieved the best AUPR and AUROC on GeneSPIDER benchmarks at moderate and low noise levels (SNR = 1 and SNR = 0.1), (PubMed Central) and the highest AUROC on 3 of 5 DREAM4 networks. (PubMed Central) Its inferred density distributions most accurately matched true GRN densities compared to LASSO, GENIE3, and LSCON. (PubMed Central) (Diva-portal) Correctly inferred links showed narrow, peaked posteriors; absent links showed broad posteriors centered near zero.

The critical differences from B-JEPA illuminate what B-JEPA adds. First, BiGSM operates directly in observed expression space without latent representation learning — it cannot capture nonlinear regulatory patterns that a JEPA encoder would extract. Second, BiGSM uses SBL/ARD Gaussian priors rather than horseshoe priors; the horseshoe's half-Cauchy hierarchy provides stronger shrinkage of near-zero effects while preserving large signals, and produces better-calibrated credible intervals than SBL priors (as established by the Cluster 3 literature). Third, BiGSM requires a known perturbation design matrix for optimal performance (nih) — a constraint B-JEPA avoids. Fourth, BiGSM's linear model cannot capture the nonlinear regulatory dynamics that B-JEPA's deep predictor handles.

GENIE3 remains the gold-standard frequentist baseline after fifteen years

Huynh-Thu et al. (2010) introduced GENIE3, which decomposes the p-gene network inference problem into **p independent regression subproblems**: for each target gene j , a Random Forest (or Extra-Trees ensemble) is trained to predict gene j 's expression from all other genes, with feature importance scores serving as edge weight proxies. (PubMed) The feature importance — total reduction in variance attributable to splits on a candidate regulator, averaged across all trees — provides a natural ranking of regulatory strength.

GENIE3 won the DREAM4 In Silico Multifactorial Challenge and consistently ranked among the top individual methods in DREAM5. It handles nonlinear interactions naturally (tree-based models capture these implicitly), produces directed graphs (each regression has an explicit target), allows feedback loops (no DAG constraint), and scales to genome-wide networks (~7 **hours** for 4,500 genes with 300 TFs). (PLOS) Using a known TF list substantially improves performance by restricting the feature space to biologically plausible regulators.

GENIE3 serves as B-JEPA's **primary Experiment 1 baseline** — the most credible frequentist method to outperform. Its limitations define B-JEPA's value proposition: no uncertainty quantification (feature importance scores are relative rankings without probabilistic interpretation), no latent state representation (operates directly on raw expression), no explicit sparsity modeling (relies on implicit feature selection), and no calibrated confidence intervals for individual edge predictions. B-JEPA must match or exceed GENIE3's AUROC/AUPR while providing the calibrated uncertainty that GENIE3 cannot.

LatentDAG and GRNVBEM address latent structure and Bayesian time-series inference respectively

Gao and Gerstein (2024) developed LatentDAG, which applies Bayesian network structure learning to gene expression data to extract a simplified "core" regulatory architecture. The resulting DAG is **two orders of magnitude simpler** than conventional co-expression or ChIP-seq networks, with clear module separators (Oxford Academic) and strong intra-module expression correlation (permutation test $P < 0.0004$). (PubMed Central) When paired with Graph Neural Networks, LatentDAG outperformed ChIP-GRN, PPI, and co-expression networks on downstream tasks including gene conservation prediction and clustering. (Oxford Academic) LatentDAG demonstrates that latent regulatory structure is biologically meaningful and computationally useful, but it does not combine this with prediction architectures or provide per-edge uncertainty. B-JEPA advances this idea by making latent structure both predictive (via JEPA) and probabilistic (via horseshoe posteriors).

Sanchez-Castillo et al. (2018) established the variational Bayesian framework for GRN inference using a first-order autoregressive moving-average model (AR1MA1) with VBEM inference. The method handles time-series and pseudo-temporally ordered single-cell data, (oup) recovering known regulatory interactions in mouse embryonic cells and zebrafish hematopoiesis. (PubMed) (Oxford Academic) However, it is **restricted to time-labeled data**, uses no sparsity-inducing priors, (Diva-portal) assumes linear dynamics, and was validated only on small networks (~20–50 genes). (PubMed Central) B-JEPA addresses all these limitations: it handles steady-state expression data, incorporates horseshoe priors for sparsity, learns nonlinear representations via JEPA, and targets genome-scale networks.

Synthesis: the GRN inference landscape and B-JEPA's position

The field divides into frequentist methods (GENIE3, TIGRESS, LASSO-based approaches) that scale well and produce reasonable point estimates but lack uncertainty quantification, and

Bayesian methods (BiGSM, GRNVBEM, classical Bayesian networks) that provide some probabilistic framework but have not incorporated modern representation learning. No existing method combines latent representation learning, Bayesian sparse inference with heavy-tailed priors, and calibrated uncertainty quantification. (PubMed Central) BiGSM comes closest but operates in observed space with SBL priors; LatentDAG captures latent structure but without predictive inference or UQ; GENIE3 provides strong empirical performance but no posteriors whatsoever.

Cluster 3: The horseshoe prior and why it is the right statistical machinery for B-JEPA

The horseshoe's U-shaped shrinkage profile is uniquely suited to sparse biological networks

Carvalho, Polson, and Scott (2010) introduced the horseshoe estimator for the sparse normal-means problem, (Oxford Academic) constructing a hierarchical prior where $\beta_j | \lambda_j, \tau \sim N(0, \lambda_j^2 \tau^2)$ with local scales $\lambda_j \sim C^+(0, 1)$ and global scale $\tau \sim C^+(0, 1)$, both following half-Cauchy distributions. (fsu) The defining property is that the shrinkage coefficient $\kappa_j = 1/(1 + \lambda_j^2)$ follows a **Beta(1/2, 1/2) distribution** — the horseshoe shape that gives the prior its name. (fsu) This distribution has infinite density (poles) at both $\kappa = 0$ and $\kappa = 1$, creating a sharp phase transition: coefficients are either shrunk almost entirely to zero or left essentially unshrunk. (fsu)

This U-shaped profile contrasts sharply with the LASSO's Laplace prior, which implies a κ distribution that is bounded at $\kappa = 1$ (weak noise shrinkage) and vanishes at $\kappa = 0$ (overshrinks signals). (fsu) The horseshoe achieves bounded influence (Fsu) — $|y - E(\theta|y)| \leq b_\tau$ for all y — meaning large signals are left approximately at their observed values. (fsu) It also achieves **super-efficient convergence** for zero coefficients (Oxford Academic) at rate $O(n^{-1}(\log n - b \log \log n))$, faster than any prior with bounded density at the origin. (fsu)

For GRN inference where each gene's expression is regressed on ~1,500 TF expression levels but only **5–20 TFs** are true regulators, the horseshoe's properties are ideal. The ~1,480 zero TF→gene connections are aggressively shrunk to zero (pole at $\kappa = 1$), while the ~20 true regulatory coefficients retain their full, unbiased effect sizes (pole at $\kappa = 0$). No other continuous shrinkage prior achieves this simultaneous property with the horseshoe's theoretical guarantees.

The regularized horseshoe solves practical implementation challenges critical for genome-scale inference

Piironen and Vehtari (2017) addressed two practical limitations of the original horseshoe that would otherwise prevent its application at genome scale. (arXiv) The first problem: no principled way to set the global shrinkage parameter τ based on prior sparsity knowledge. (arXiv) (arxiv) Their solution is the **τ_0 formula**:

$$\tau_0 = (p_0 / (D - p_0)) \cdot (\sigma / \sqrt{n})$$

where p_0 is the expected number of nonzero coefficients, D is the total number of predictors, σ is the noise scale, and n is the sample size. This formula centers the prior on the effective number of nonzero parameters m_{eff} at p_0 , ensuring consistency as n and D change. (Statistical Modeling) (arxiv)

The second problem: the original horseshoe has unbounded tails (infinite slab width), causing divergent transitions in Hamiltonian Monte Carlo sampling and instability with weakly identified parameters. (arxiv) The regularized (Finnish) horseshoe replaces the local scale λ_j with $\tilde{\lambda}_j^2 = c^2 \lambda_j^2 / (c^2 + \tau^2 \lambda_j^2)$, where c is a finite slab width. (Statistical Modeling) When λ_j is small (noise), $\tilde{\lambda}_j \approx \lambda_j$ (unchanged shrinkage); when λ_j is large (signal), $\tilde{\lambda}_j \rightarrow c/\tau$ (bounded). The prior on unshrunk coefficients approaches $N(0, c^2)$ — a finite-width slab that prevents extreme values while preserving the horseshoe's essential U-shaped spike component. The slab scale $c^2 \sim \text{Inv-Gamma}(v_{\text{slab}}/2, v_{\text{slab}} \cdot s^2_{\text{slab}}/2)$ receives its own prior, (arxiv) with typical defaults $v_{\text{slab}} = 4$, $s_{\text{slab}} = 2$. (Jrnl)

This is **directly implemented in B-JEPA**. For each target gene g with expression $y_g \in \mathbb{R}^n$ regressed on TF expression $X \in \mathbb{R}^{n \times D}$: $\beta_{\{g,j\}} \mid \tilde{\lambda}_j, \tau \sim N(0, \tau^2 \tilde{\lambda}_j^2)$, with $\lambda_j \sim C^+(0,1)$, $\tau \sim \text{Half-}C^+(0, \tau_0)$, and $c^2 \sim \text{Inv-Gamma}$. Setting $p_0 \approx 10\text{--}20$ (the typical number of regulators per gene) and $D \approx 334$ (the number of TFs in the DREAM5 E. coli dataset) yields a principled, data-informed prior that encodes biological sparsity knowledge.

LASSO posteriors are demonstrably unsuitable for the uncertainty quantification B-JEPA requires

Bhadra et al. (2019) provided a comprehensive survey establishing why horseshoe-based inference is strictly preferable to LASSO for uncertainty quantification (Project Euclid) — the central statistical argument underlying B-JEPA's design. The critical finding, drawing on Castillo, Schmidt-Hieber, and van der Vaart (2015), is devastating for LASSO-based UQ: **"The LASSO is essentially non-Bayesian, in that the full posterior distribution is useless for uncertainty quantification."** (Project Euclid) (arxiv) Specifically, while the LASSO point estimate (posterior mode under the Laplace prior) achieves near-minimax risk, the full posterior contracts at a much slower rate than the mode. (arXiv) Credible intervals derived from the Bayesian LASSO posterior do not have correct frequentist coverage, and the posterior mean does not inherit the mode's optimality properties.

By contrast, the horseshoe posterior contracts at the near-minimax rate around the true parameter, (arXiv) and horseshoe credible intervals provide asymptotically calibrated coverage. (Scholarly Publications) The shrinkage profile comparison is instructive: the horseshoe leaves large inputs unshrunk ($E(\theta|y) \approx y$ for large $|y|$) and shrinks small inputs to zero, (Fsu) while the LASSO shrinks all inputs by a non-vanishing amount, introducing systematic bias for true signals.

For B-JEPA, this paper provides the primary argument against point-estimate GRN methods. GENIE3, GRNBoost2, and other frequentist methods cannot provide reliable uncertainty over

network edges — they produce rankings without calibrated probabilities. LASSO-based approaches (TIGRESS, LSCON) might seem to offer a Bayesian alternative, but Bhadra et al. demonstrate that their posteriors are fundamentally unreliable for UQ. Only horseshoe-type priors provide the theoretical backing for calibrated credible intervals that B-JEPA promises.

The horseshoe achieves near-optimal Bayes risk for sparse signal detection

Datta and Ghosh (2013) proved that the horseshoe's implicit testing rule — rejecting $H_0: \theta_i = 0$ when the shrinkage weight $(1 - \hat{\kappa}_i) > 1/2$ — is **Asymptotically Bayes Optimal under Sparsity (ABOS)**. [\(Project Euclid\)](#) Under the two-groups normal model with growing dimension m , increasing sparsity ($p_m \rightarrow 0$), and signals near the detection boundary ($\psi_m^2 \sim 2 \log(1/p_m)$), the horseshoe Bayes risk satisfies $R_{HS} = O(R_{opt})$, matching the oracle risk up to a multiplicative constant. [\(Project Euclid\)](#) The proof demonstrates that κ_i concentrates near 1 for noise observations (full shrinkage) and near 0 for signals (no shrinkage), with type I and type II error rates tracking the Bayes oracle closely. [\(Project Euclid\)](#)

Numerical simulations with $m = 200$ confirmed that the horseshoe decision rule's misclassification probability tracks the oracle at all tested sparsity levels, substantially outperforming the LASSO decision rule. [\(Project Euclid\)](#) Full Bayes estimation of τ (with half-Cauchy prior) adapts well to unknown sparsity, [\(arXiv\)](#) while the empirical Bayes estimate tends to collapse toward zero. [\(Project Euclid\)](#) This theoretical guarantee validates B-JEPA's claim that horseshoe-based edge detection in GRN inference operates at a near-optimal statistical efficiency — the cost of using a continuous prior rather than the intractable spike-and-slab is provably negligible.

Cluster 4: Predictive coding and the Bayesian brain as B-JEPA's intellectual lineage

Hierarchical predictive coding provides the computational principle B-JEPA inherits

Rao and Ballard (1999) proposed that the visual cortex implements hierarchical predictive coding: higher cortical areas generate top-down predictions of activity at lower areas, while lower areas compute prediction errors and propagate them upward. [\(Nature\)](#) [\(Washington\)](#) The mathematical formulation is a gradient descent on the negative log-posterior: $dr/dt = (k_i/\sigma^2) \cdot U^T(I - f(Ur)) + (k_i/\sigma_{td}^2)(r_{td} - r) - (k_i/2)g'(r)$. The estimate r is updated using bottom-up error weighted by inverse noise variance $1/\sigma^2$, top-down error weighted by $1/\sigma_{td}^2$, and a prior penalty $g'(r)$. **The weight given to each information source is inversely proportional to its noise variance** — a precision-weighting principle central to Bayesian inference. [\(Washington\)](#)

A three-level network trained on natural images developed Gabor-like oriented edge detectors at level 1 (resembling V1 simple cells) and explained extra-classical receptive field effects: error-detecting neurons showed strong responses to short bars (unpredictable from context) but weak responses when bars extended into the surround (predictable). [\(ResearchGate\)](#) Removing top-

down feedback eliminated end-stopping in **82%** of model neurons, paralleling experimental data from cat V1. (washington) The model produced **85.3% response suppression** with iso-oriented surrounds and 19.1% facilitation with cross-oriented surrounds, matching neurophysiological observations. (washington)

The connection to B-JEPA is structural: JEPA's predictor module is analogous to the top-down generative model (backward connections), the encoder produces representations analogous to processed sensory signals, and the prediction loss in latent space mirrors the precision-weighted prediction error. (Substack) B-JEPA's horseshoe prior on regulatory edges is analogous to the structural prior on connectivity patterns, and the posterior over edge weights corresponds to updated beliefs about the causes of observed gene expression.

Bayesian brain theory bridges JEPA world models and Bayesian inference

Bottemanne et al. (2025) synthesized Bayesian Brain Theory as a computational framework where the nervous system encodes probabilistic beliefs — represented as probability densities over latent causes — within neural networks. (ScienceDirect) (Neuroscience) The core thesis is that the brain combines prior beliefs with sensory evidence via Bayes' rule: $P(\text{hypothesis} | \text{data}) \propto P(\text{data} | \text{hypothesis}) \times P(\text{hypothesis})$. (ResearchGate) Critically, the brain does not treat all prediction errors equally: each is weighted by its **precision** (inverse variance), determining the relative influence of bottom-up evidence versus top-down predictions. (Neuroscience) High precision on sensory data leads to strong belief updating; high precision on priors leads to resistance to updating. Precision is encoded by neuromodulators including dopamine, serotonin, and norepinephrine, with NMDA receptor signaling encoding predictions and AMPA signaling encoding prediction errors. (Neuroscience)

This paper provides B-JEPA's **conceptual bridge**: just as the brain combines prior beliefs with sensory evidence using precision-weighted Bayesian inference, B-JEPA combines structural priors on GRN sparsity (horseshoe prior) with gene expression data (likelihood) to produce posterior beliefs about regulatory edges. The insistence that beliefs are probability distributions, not point estimates, directly supports B-JEPA's use of full posterior distributions rather than the point estimates produced by GENIE3 or other frequentist methods. The precision-weighting principle maps to the horseshoe's local shrinkage parameters λ_j , which adaptively determine how much each regulatory edge is shrunk based on data evidence.

The free energy principle provides the deepest theoretical framework for B-JEPA

Friston (2005) demonstrated that predictive coding, Bayesian brain theory, and empirical Bayes are all special cases of a universal principle: **variational free energy minimization**. (PubMed) The core formulation is $F = \langle \ln p(u; \theta) \rangle - \text{KL}[q(v; u) \parallel p(v|u; \theta)]$, where the first term is the log-evidence of sensory data under the generative model and the second is the KL divergence between the approximate and true recognition densities. Since $\text{KL} \geq 0$, F is a lower bound on log-evidence; maximizing F simultaneously improves the generative model (M-step/learning) and the

recognition density (E-step/inference). [\(nih\)](#) Under Gaussian assumptions, the free energy decomposes into precision-weighted prediction errors plus prior constraint terms. [\(Google Scholar\)](#)

The neural implementation predicts that forward connections carry prediction errors (driving, AMPA-mediated) while backward connections carry predictions (modulatory, NMDA-mediated). [\(nih\)](#) The hierarchical generative model enables empirical Bayes: the prior at level i emerges from the generative distribution at level $i+1$. [\(PubMed\)](#) Friston demonstrated that this single framework explains classical receptive fields, extra-classical surround effects, repetition suppression, mismatch [\(Royal Society Open Science\)](#) [\(Google Scholar\)](#) match negativity, and the P300 component — all as consequences of free energy minimization.

B-JEPA's loss function can be formally recast as a variational free energy: $-F_{\text{B-JEPA}} = E_q(W) [\|s_{\text{target}} - \text{Pred}(s_{\text{context}}; W)\|^2] + \text{KL}[q(W) \parallel p_{\text{horseshoe}}(W)]$, where the first term is expected prediction error in latent space and the second is the divergence between the approximate posterior over regulatory edges and the horseshoe prior. This framing places B-JEPA within a precise theoretical position: it applies the free energy principle to gene regulatory network inference, using JEPA's latent-space prediction as the computational substrate. The brain minimizes free energy to infer causes of its sensory world; B-JEPA minimizes free energy to infer the regulatory structure generating observed gene expression.

Cluster 5: The virtual cell paradigm and Perturb-seq validation datasets

The virtual cell roadmap explicitly calls for the uncertainty B-JEPA provides

Bunne et al. (2024), a perspective paper from 42 co-authors spanning Stanford, C [\(Cell Press\)](#) ZI, Genentech, HHMI, Google, and Microsoft Research, articulated the vision for an AI Virtual Cell — a multi-scale, multi-modal neural network that represents and simulates molecular, cellular, and tissue behavior. The paper identifies gene regulation as a priority area and envisions "virtual instruments" that simulate perturbations and predict cell state transitions. The paper explicitly calls for Bayesian formulations: **"Under a Bayesian formulation of its predictive function, the predictions made for cell perturbation outcomes could include an uncertainty score, either implicitly via inference, deep kernels, or through explicit estimation of the full posterior over model parameters."** The paper also envisions active learning and lab-in-the-loop schemes guided by uncertainty estimates.

B-JEPA directly answers this call. It provides the uncertainty-aware GRN inference module that the virtual cell paradigm requires but no existing foundation model delivers. Its calibrated posterior distributions over regulatory edges enable exactly the confidence-weighted experimental prioritization that Bunne et al. envision.

CZI's foundation models achieve impressive scale but lack calibrated regulatory uncertainty

CZI's virtual cell ecosystem now includes three major models. **TranscriptFormer** (Pearce et al., 2025) is a transformer encoder trained on **112 million cells from 12 species** spanning 1.5 billion years of evolution, with ~300M trainable parameters. It achieves macro F1 up to 0.91 on cell type classification and enables cross-species annotation even at 600+ million years evolutionary distance. **GREmLN** (Zhang et al., 2025) integrates gene regulatory network structure directly into its attention mechanism via graph signal processing, achieving state-of-the-art with only **10.3 million parameters** — less than one-third of baselines — by leveraging biological inductive bias from ARACNe-inferred GRNs. **rBio** (2025) is a reasoning LLM trained on virtual cell model outputs using reinforcement learning, enabling plain-language biological queries.

The critical gap across all three models is the absence of calibrated uncertainty over regulatory predictions. TranscriptFormer produces point estimates of gene expression without confidence intervals for specific TF→target relationships. GREmLN treats regulatory graphs as **static, fixed inputs** — "the graph structure needs to remain static during training and evaluation" — inheriting all errors from the upstream inference method without the ability to express uncertainty about individual edges. rBio operates at the natural language level, not at the molecular level required for calibrated regulatory inference. None provides a principled Bayesian posterior over regulatory structures. B-JEPA fills this gap: its calibrated regulatory posteriors could serve as higher-quality, uncertainty-weighted GRN inputs for GREmLN, or as an uncertainty layer on TranscriptFormer representations.

Perturb-seq datasets provide the causal ground truth for validating B-JEPA's predictions

Norman et al. (2019) constructed genetic interaction manifolds from combinatorial CRISPRa perturbations in K562 cells, profiling **287 perturbations** (105 single-gene, 131 two-gene combinations) across **~110,000 single cells**. The GI manifold revealed that scalar fitness measurements collapse a much richer landscape — distinct interactions with similar fitness effects occupied markedly different transcriptional states. The dataset discovered unexpected synergies (CBL and CNN1 driving erythroid differentiation) invisible in fitness-only data. This processed dataset has become the **standard benchmark** for perturbation prediction models including GEARS, scGPT, and scFoundation.

Replogle et al. (2022) scaled Perturb-seq to the **entire expressed genome**, targeting **9,866 genes** via CRISPRi across **>2.5 million K562 cells**, plus 2,393 essential genes in RPE1 cells. The study used transcriptional phenotypes to predict functions of uncharacterized genes, identifying CCDC86, ZNF236, and SPATA5L1 as ribosome biogenesis regulators and C7orf26 (renamed INTS15) as a new Integrator complex subunit. With **7,189 genes** showing differentially expressed signatures, this dataset provides genome-wide causal evidence for regulatory relationships.

Together, these datasets constitute B-JEPA's Phase 2 validation framework. Norman provides combinatorial gain-of-function data testing whether B-JEPA's latent regulatory structure

captures real genetic interactions. Replogle provides genome-wide loss-of-function data enabling systematic assessment of whether B-JEPA's uncertainty-calibrated TF→target predictions are validated by causal perturbation evidence — and critically, whether the 95% credible intervals actually contain the true perturbation effects 95% of the time. The complementarity — gain versus loss of function, combinatorial versus single-gene, moderate versus genome-wide scale — provides rigorous multi-angle validation.

Novelty analysis: what is genuinely new in B-JEPA

B-JEPA's novelty lies not in any single component but in a **specific five-way integration** that does not exist in any paper or obvious two-paper combination:

The first novel element is **JEPA applied to biological network data**. All existing JEPA work operates on images or video in computer vision. Adapting the context-encoder / predictor / target-encoder architecture to TF expression → target gene expression prediction is without precedent. The "masking" in B-JEPA is not artificial (randomly dropping patches) but reflects the genuine information asymmetry of gene regulation.

The second is **horseshoe priors within a JEPA architecture**. No JEPA variant incorporates Bayesian inference of any kind. The latent variable z in LeCun's framework is deterministic or learned via amortized inference; B-JEPA replaces this with a full horseshoe posterior over regulatory weights, creating a fundamentally probabilistic JEPA.

The third is the **combination of latent representation learning with sparse Bayesian regression for GRN inference**. BiGSM performs Bayesian sparse regression but in observed space. LatentDAG learns latent structure but without Bayesian edge-level inference. GENIE3 provides strong performance without either. No method combines a deep encoder that learns latent gene expression representations with horseshoe-prior-based sparse regression in that latent space.

The fourth is **calibrated per-edge uncertainty quantification backed by optimal decision theory**. The horseshoe's ABOS property (Datta and Ghosh, 2013), combined with its demonstrated superiority over LASSO for UQ (Bhadra et al., 2019), provides theoretical backing that no existing GRN method offers. B-JEPA's credible intervals are not ad hoc confidence scores but posteriors with proven asymptotic calibration.

The fifth is the **free energy / predictive coding framing applied to gene regulation**. Connecting GRN inference to the Bayesian brain literature through the free energy principle provides both a theoretical justification for the architecture and a broader narrative about how biological systems — both neural circuits and gene regulatory networks — can be understood as minimizing surprise through predictive models.

Gap analysis: what the field lacks that B-JEPA provides

Gap 1: No existing GRN inference method simultaneously learns latent representations, performs Bayesian sparse inference, and produces calibrated uncertainty. Frequentist methods (GENIE3, TIGRESS) produce point estimates. Bayesian methods (BiGSM, GRNVBEM) operate in observed space. Latent methods (LatentDAG) lack per-edge UQ. B-JEPA is the first to combine all three capabilities.

Gap 2: Foundation models for biology lack calibrated uncertainty over regulatory predictions. TranscriptFormer, GREmLN, and scGPT produce point estimates of expression or treat GRN structure as fixed input. The virtual cell roadmap (Bunne et al., 2024) explicitly identifies this gap and calls for Bayesian formulations. B-JEPA provides the missing uncertainty-aware GRN component.

Gap 3: The JEPA paradigm has never been applied outside computer vision. Despite its theoretical elegance and empirical success on images and video, no work has adapted JEPA to structured biological prediction tasks where the "masking" corresponds to genuine causal asymmetry rather than artificial patch removal.

Gap 4: No bridge exists between the Bayesian brain / predictive coding literature and practical gene regulatory inference. The intellectual connection between how the brain infers causes of sensory input and how we might infer causes of gene expression patterns has been noted informally but never operationalized in a working method. B-JEPA is this bridge.

Gap 5: Existing Bayesian GRN methods use suboptimal priors for sparse biological networks. BiGSM uses SBL/ARD Gaussian priors; GRNVBEM uses standard conjugate priors. Neither exploits the horseshoe's theoretically optimal properties for sparse signal recovery, super-efficient convergence for null effects, or ABOS-optimal testing. B-JEPA is the first GRN method to deploy the regularized horseshoe with the principled τ_0 formula encoding biological sparsity information.

Positioning statement

B-JEPA occupies a unique position at the intersection of three mature but previously disconnected fields: the JEPA paradigm for self-supervised latent prediction (LeCun, 2022; Assran et al., 2023), Bayesian sparse inference via horseshoe priors with provable optimality guarantees (Carvalho et al., 2010; Piironen and Vehtari, 2017), and gene regulatory network reconstruction from expression data (Marbach et al., 2012; Huynh-Thu et al., 2010). While existing GRN methods either provide strong point estimates without uncertainty (GENIE3) or Bayesian posteriors without latent representation learning (BiGSM), and while foundation models for biology produce impressive cell-level predictions without calibrated regulatory

uncertainty (TranscriptFormer, GREmLN), B-JEPA is the first method to combine JEPA-style prediction in learned latent space with horseshoe-prior-based sparse Bayesian inference, yielding per-edge posterior distributions over regulatory relationships that are backed by near-minimax optimality theory and validatable against genome-wide Perturb-seq causal ground truth.